Type MS · Proving identities with double-angle and addition formulae

Answer 1 of 14 · Question 9 part b · 5 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
9(a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A) = 2\cos^2 A - 1$ OR $\cos 2A = \cos^2 A - \sin^2 A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ (i) $\cos^2 A = \frac{\cos 2A + 1}{2}$, (ii) $\sin^2 A = \frac{1-\cos 2A}{2}$ *	M1M1 A1A1cso [4]
(b)	$(2\sin x - \cos x)(\sin x - 3\cos x)$ $= 2\sin^2 x - 7\sin x\cos x + 3\cos^2 x$ $= 2\left(\frac{1-\cos 2x}{2}\right), \ldots\ldots\ldots\ldots, +3\left(\frac{\cos 2x+1}{2}\right)$ $= \ldots\ldots\ldots\ldots - \frac{7\sin 2x}{2} + \ldots\ldots\ldots\ldots$ $= \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$*	M1 M1, M1 M1 A1 cso [5]
(c)	$y = \frac{1}{2}(\cos 2x - 7\sin 2x + 5) \Rightarrow \frac{dy}{dx} = -\sin 2x - 7\cos 2x = 0$ $\Rightarrow -\sin 2x = 7\cos 2x \Rightarrow \tan 2x = -7$ $2x = -81.869, \quad 98.130, \quad 278.130$ $\Rightarrow x = 49, \quad 139$	M1 M1 A1 A1 [4]
		Total 13 marks

Part	Mark	Notes
NB: Parts (a) and (b) are proofs with given answers. You must check every line of their working.		
(a)	M1	Uses the formula for $\cos(A+B) = \cos A\cos B - \sin A\sin B$ correctly.
	M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$
(i)	A1 cso	For the correct identity for either $\sin^2 A$ or $\cos^2 A$ with no errors seen
(ii)	A1 cso	For the correct identity for $\sin^2 A$ and $\cos^2 A$ with no errors seen
SC	If a candidate does not use $\cos(A+B) = \cos A\cos B - \sin A\sin B$ to find the required identities for $\sin^2 A$ or $\cos^2 A$ but uses them to show $\sin^2 A = \frac{1-\cos 2A}{2}$ or $\cos^2 A = \frac{\cos 2A+1}{2}$ apply the following scheme.	
	First M mark – M0	
	2nd M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$ For example:

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		$\cos^2 A = \frac{\cos 2A + 1}{2} = 1 - \sin^2 A$ $\sin^2 A = 1 - \frac{\cos 2A + 1}{2} = \frac{2 - (\cos 2A + 1)}{2} = \frac{1 - \cos 2A}{2}$
(i)/(ii)	**A1 cso**	For the correct identity for $\sin^2 A$ or $\cos^2 A$ with no errors seen
	A0	Not available
(b)	**M1**	Multiplies out the brackets with no errors.
	M1	Substitutes at least one of the given identities for $\cos 2x$ into their expansion [either for $\sin^2 x$ or $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression.
	M1	Substitutes both of the given identities for $\cos 2x$ into their expansion [for both $\sin^2 x$ and $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression. **NB: This is a B mark in Epen**
	M1	Uses the identity $\sin 2A = 2\sin A \cos A$ seen applied explicitly **before** the final answer. Look for, $k \sin x \cos x \rightarrow \frac{k \sin 2x}{2}$
	A1 cso	For the correct identity with no errors seen.
	ALT – Works backwards – these steps must contain no errors (to maintain compatibility with the main method. Starting from $\frac{1}{2}(\cos 2x - 7\sin 2x + 5)$ award as follows:	
	M1	Uses: $\cos 2x = \cos^2 x - \sin^2 x$
	M1	Uses: $5 = 5\cos^2 x + 5\sin^2 x$
	M1	Uses: $-7\sin 2x = 14 \sin x \cos x$
	M1	Collects like terms and divides by 2 to give: $3\cos^2 x - 7\sin x \cos x + 2\sin^2 x$
	A1	Factorises their expression to give: $(2\sin x - \cos x)(\sin x - 3\cos x)$
(c)	**M1**	For an attempt to differentiate the given equation. A minimally acceptable is defined as follows: $\frac{dy}{dx} = \pm k \sin 2x \pm l \cos 2x$ where k and l are integers **with no other terms**.
	M1	Uses the identity $\tan 2x = \frac{\sin 2x}{\cos 2x}$ This is an independent mark, but it must be used correctly following correct division of $\cos 2x$ throughout.
	A1	Find at least one **correct** value for $2x$ e.g., –81.869, 98.130, 278.130 etc **NB: This is an M mark in Epen.** [Accept – 81, 98, 278]
	A1	For $x = 49$ and 139 rounded to the nearest whole number or better.

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	ALT - Differentiates using product rule	
	M1	$y=(2\sin x-\cos x)(\sin x-3\cos x)$ $\frac{dy}{dx}=(2\sin x-\cos x)(\cos x+3\sin x)+(\sin x-3\cos x)(2\cos x+\sin x)$ $\frac{dy}{dx}=2\sin x\cos x+6\sin^2 x-\cos^2 x-3\sin x\cos x+$ $+2\sin x\cos x+\sin^2 x-6\cos^2 x-3\sin x\cos x$ $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x$ **Allow** $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ **NB: This will not factorise**
	M1	Divides through by $\cos^2 x$ provided they have an expression of the form $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ This gives : $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x=0 \quad \rightarrow \div\cos^2 x$ $-7-2\tan x+7\tan^2 x=0$ **AND** solves to give a value for $\tan x=\frac{1\mp 5\sqrt{2}}{7}$ or allow $\tan x=\frac{1\pm 5\sqrt{2}}{7}$
	A1	At least one correct value for x of for example, awrt 49.1….awrt – 40.9
	A1	For both 49° and 139°

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Question	Scheme	Marks

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Answer 2 of 14 · Question 10 part b · 4 marks

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Number		
10 (a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - \left(1 - \cos^2 A\right) = 2\cos^2 A - 1*$	M1 A1cso [2]
(b)	$\cos 4A = 2\cos^2 2A - 1$ $= 2\left(2\cos^2 A - 1\right)^2 - 1$ $= 2\left(4\cos^4 A - 4\cos^2 A + 1\right) - 1$ $= 8\cos^4 A - 8\cos^2 A + 1*$	M1 M1 M1 A1cso [4]
(c)	Let $\frac{\theta}{4} + \frac{\pi}{24} = x$ $\cos^2 x\left(\cos^2 x - 1\right) = \cos^4 x - \cos^2 x$ $\Rightarrow \cos^4 x - \cos^2 x = \frac{\cos 4x - 1}{8}$ $4\left(\frac{\theta}{4} + \frac{\pi}{24}\right) = \theta + \frac{\pi}{6}$ $\frac{\cos\left(\theta + \frac{\pi}{6}\right) - 1}{8} = -\frac{1}{16} \Rightarrow \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$ $\Rightarrow \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 A1 [5]
Alternative (with x defined as above)		
(c)	$\cos^4 x - \cos^2 x = -\frac{1}{16} \Rightarrow 8\cos^4 x - 8\cos^2 x + 1 = 1 - \frac{8}{16}$ $\cos 4x = \frac{1}{2} \Rightarrow 4x = \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \cdots \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1M1 M1A1
Alternative (with x defined as above)		
(c)	$16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0$ $\cos^2 x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4\times16\times1}}{2\times16} = \frac{2 \pm \sqrt{3}}{4}$ $\cos x = \sqrt{\frac{2 \pm \sqrt{3}}{4}}$ $(= \pm 0.9659\ldots \text{ or } \pm 0.2588\ldots)$ $x = 0.2617\ldots, 1.3089\ldots, 1.8325\ldots, 2.8787\ldots \left(= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}\ldots\right)$ $\theta = 4\left(x - \frac{\pi}{24}\right) = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 M1A1
(d)	$\mathrm{f}(A) = \frac{\cos 4A}{2} + \frac{1}{2}$ $\left[\frac{\sin 4A}{8} + \frac{A}{2}\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \left(\frac{\sin 2\pi}{8} + \frac{\pi}{4}\right) - \left(\frac{\sin\frac{2\pi}{3}}{8} + \frac{\pi}{12}\right) = \frac{\pi}{4} - \frac{\sqrt{3}}{16} - \frac{\pi}{12} = \frac{\pi}{6} - \frac{\sqrt{3}}{16}$	M1 dM1 ddM1 A1 [4]
		Total 15 marks

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Answer 2 of 14 · Question 10 part b · 4 marks

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(a)	M1	Use $\cos(A+A)=\cos A\cos A-\sin A\sin A$ (may go straight to $\cos 2A=\cos^2 A-\sin^2 A$) and then try to replace $\sin^2 A$ by $1-\cos^2 A$ Allow with any other single variable.
	A1 cso	Obtain $\cos 2A=2\cos^2 A-1$ with no errors in the working. Must be stated with A as the variable.
(b)		Allow with any other single variable for all M marks. Allow one sign error removing brackets and ft for all M marks.
	M1	$\cos 4A=2\cos^2 2A-1$ or $\cos 4A=\cos^2 2A-\sin^2 2A$ Obtain an expression in terms of cos 2A and/or sin 2A.
	M1	eg $\cos 4A=2\left(2\cos^2 A-1\right)^2-1$, $\cos 4A=\left(2\cos^2 A-1\right)^2-4\left(1-\cos^2 A\right)\cos^2 A$ Obtain a correct expression in cos A. Eliminate terms in sin A and sin 2A.
	M1	Expand brackets. Numerical factors may remain. e.g. $\cos 4A=2\left(4\cos^4 A-2\cos^2 A-2\cos^2 A+1\right)-1$ e.g. $\cos 4A=4\cos^4 A-4\cos^2 A+1-4\cos^2 A+4\cos^4 A$
	A1 cso	Obtain the given result with no errors in the working. Must be stated with A as the variable.
(c)	M1	Expand brackets and use the result from part (b) to express the LHS of the equation in terms of $4x$, $\frac{\cos 4x-1}{8}$, where $x=\frac{\theta}{4}+\frac{\pi}{24}$ **Alternative** Arrange the equation in the form $8\cos^4 x-8\cos^2 x+1=1-\frac{8}{16}$ Allow a single numerical or sign error in the constants for both methods.
	M1	Rearrange the equation to give $\cos\left(\theta+\frac{\pi}{6}\right)=\frac{1}{2}$ or oe Note that this mark is shown as B1 on ePEN.
	M1	Find any correct positive value for $4\left(\frac{\theta}{4}+\frac{\pi}{24}\right)$, given in radians. eg $\frac{\pi}{3}$
	M1	Find any correct positive value for θ, given in radians. eg $\frac{\pi}{6}$
	A1	Both correct answers with no extras, given in radians as multiples of π. Ignore answers outside the given range.

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Alternative		
(c)	M1	Write the equation as a three term quadratic. e.g. $16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0, \left(x = \frac{\theta}{4} + \frac{\pi}{24}\right)$, $A^2 - A + \frac{1}{16} = 0, \left(A = \cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\right)$
	M1	Any valid method to solve their quadratic equation. Note that this mark is shown as B1 on ePEN.
	M1	Any correct value for $x = \frac{\theta}{4} + \frac{\pi}{24}$ in the range $0 \le x \le \pi$, given in radians, either as a decimal or as a multiple of π.
	M1	Either value of θ given in radians, $\theta = \frac{\pi}{6}$ or $\theta = \frac{3\pi}{2}$ or $\theta = 0.523...$ or $\theta = 4.712...$
	A1	Both $\theta = \frac{\pi}{6}$ and $\theta = \frac{3\pi}{2}$ given as multiples of π.
(d)	M1	Use result from part (b) to write $\mathrm{f}(A) = k(\cos 4A \pm 1)$.
	dM1	Integrate their $\mathrm{f}(A)$ correctly, dep on M1 awarded. Ignore limits.
	ddM1	Substitute both given limits, dep on both M marks awarded.
	A1	$\frac{\pi}{6} - \frac{\sqrt{3}}{16}$ *or* $\frac{1}{6}\pi - \frac{1}{16}\sqrt{3}$ simplified in this format. Do not isw.

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Answer 2 of 14 · Question 10 part b · 4 marks

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Question number	Scheme	Marks

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Answer 3 of 14 · Question 7 part a · 4 marks

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7 (a)	$\dfrac{3+\left(1-\cos^2\theta\right)}{\cos\theta-2}=3\cos\theta$	M1
	$4-\cos^2\theta=3\cos^2\theta-6\cos\theta$	M1
	$(2\cos\theta+1)(2\cos\theta-4)=0$	M1
	$(\cos\theta=2$ does not exist so) $\cos\theta=-\dfrac{1}{2}$ *	A1 cso (4)
(b)	$\cos 3x=-\dfrac{1}{2}$	M1
	$3x=120^\circ,\ 240^\circ,\ 480^\circ$	A1
	$x=40^\circ,\ 80^\circ,\ 160^\circ$	A1 A1 (4)
		Total 8 marks

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Answer 3 of 14 · Question 7 part a · 4 marks

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		Penalise any extra angles within range by deducting the final A mark.

Question number	Scheme	Marks

Type MS · Proving identities with double-angle and addition formulae

Answer 4 of 14 · Question 11 part a · 3 marks

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Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow \frac{3}{2}\sin 2\theta,+2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow \cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow \cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow \tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	$\text{Area}=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

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Answer 5 of 14 · Question 10 part b · 4 marks

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10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

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11(a)(i)	$\cos 2A = \cos^2 A - \sin^2 A, = \cos^2 A - \left(1 - \cos^2 A\right)$ $\cos 2A = 2\cos^2 A - 1$ *	M1,M1 A1 cso [3]
(ii)	$\sin 2A = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$ *	B1 cso [1]
(b)	$\cos 3A = \cos\left(2A + A\right)$ $= \cos 2A\cos A - \sin A\sin 2A$ $= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A$ $= 2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A$ $= 2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A$ $= 4\cos^3 A - 3\cos A$ $\Rightarrow \cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ * **ALT** $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ $= \dfrac{\cos 2A\cos A - \sin 2A\sin A + 3\cos A}{4}$ $= \dfrac{\left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A + 3\cos A}{4}$ $= \dfrac{4\cos^3 A}{4} = \cos^3 A \quad [LHS = RHS]$	M1 M1 M1 A1 Cso [4] [M1 M1 M1 A1]
(c)	$\left[\cos 3A = 4\cos^3 A - 3\cos A \Rightarrow 2\cos 3A = 8\cos^3 A - 6\cos A\right]$ $8\cos^3\left(\dfrac{\theta}{2}\right) - 6\cos\left(\dfrac{\theta}{2}\right) - 1 = 0 \Rightarrow 2\cos 3\left(\dfrac{\theta}{2}\right) - 1 = 0$ $\cos 3\left(\dfrac{\theta}{2}\right) = \dfrac{1}{2} \Rightarrow 3\left(\dfrac{\theta}{2}\right) = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, \dfrac{7\pi}{3}, \left(\dfrac{11\pi}{3}\right)$ $\theta = \dfrac{2\pi}{9}, \dfrac{10\pi}{9}, \dfrac{14\pi}{9}$	M1 M1 M1A1 [4]

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(d)	$I = \int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta = \int_0^{\frac{\pi}{6}} \left(\cos 3\theta + 3\cos\theta - \sin 2\theta\right) \mathrm{d}\theta$	M1
	$I = \left[\frac{\sin 3\theta}{3} + 3\sin\theta + \frac{\cos 2\theta}{2}\right]_0^{\frac{\pi}{6}}$	M1
	$= \left[\frac{\sin 3\left(\frac{\pi}{6}\right)}{3} + 3\sin\left(\frac{\pi}{6}\right) + \frac{\cos 2\left(\frac{\pi}{6}\right)}{2}\right] - \left[\frac{\sin 3(0)}{3} + 3\sin(0) + \frac{\cos 2(0)}{2}\right]$	M1
	$= \left[\frac{1}{3} + \frac{3}{2} + \frac{1}{4}\right] - \left[0 - 0 + \frac{1}{2}\right] = \frac{19}{12}$	A1 [4]
		Total 16 marks

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(c)	M1	Substitutes $2\cos 3\left(\frac{\theta}{2}\right)$ in place of $8\cos^3\left(\frac{\theta}{2}\right)-6\cos\left(\frac{\theta}{2}\right)$ correctly to obtain at least $k\cos 3\left(\frac{\theta}{2}\right)-1=0$ where k is a constant
	M1	Rearranges to obtain $\cos 3\left(\frac{\theta}{2}\right)=\frac{1}{k}$ and takes the inverse cosine to find at least any one **correct** value for $3\left(\frac{\theta}{2}\right)$
	M1	Finds at least one correct value for θ
	A1	Finds all three values with no extra values within range. Ignore any values given outside of the range.
(d)	M1	For a **correct** substitution as shown. $4\cos^3\theta=\cos 3\theta+3\cos\theta$
	M1	For an acceptable attempt to integrate either $\cos 3\theta\rightarrow\pm\frac{\sin 3\theta}{3}$ or $-\sin 2\theta\rightarrow\pm\frac{\cos 2\theta}{2}$
	M1	For substituting both given values the correct way around. This mark can be awarded for a complete substitution SEEN into any changed expression. Award this mark for **correct** integration **and** a value of $\frac{19}{12}$ seen without explicit substitution.
	A1	For $\frac{19}{12}$

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8 (a)	$k = 2$	B1 [1]
(b)	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $3\cos A-\sin A-6\sin A\cos A+2\sin^2 A=3\cos A(1-2\sin A)-\sin A(1-2\sin A)$ $3\cos A(1-2\sin A)-\sin A(1-2\sin A)=(1-2\sin A)(3\cos A-\sin A)$ $\Rightarrow p=3,\ q=1,\ r=2$	B1 M1 A1 [3]
	ALT	
	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $(p\cos A-\sin A)(q-r\sin A)=pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $\Rightarrow p=3,\ q=1,\ r=2$	{B1} {M1} {A1} [3]
(c)	$(1-2\sin 2\theta)(3\cos 2\theta-\sin 2\theta)=0$ $\Rightarrow 3\cos 2\theta-\sin 2\theta=0\Rightarrow\tan 2\theta=3$ $\Rightarrow 1-2\sin 2\theta=0\Rightarrow\sin 2\theta=\frac{1}{2}$ $\tan 2\theta=3\Rightarrow 2\theta=1.2490...,\ 4.3906...\ \Rightarrow\theta=0.625,\ 2.20$ $\sin 2\theta=\frac{1}{2}\Rightarrow 2\theta=0.5235...,\ 2.6179...\ \Rightarrow\theta=0.262,\ 1.31\ \left[\frac{\pi}{12},\frac{5\pi}{12}\right]$	B1 B1 M1A1 M1A1 [6]
		Total 10 marks

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(c)	
B1	$\tan 2\theta = 3$
B1	$\sin 2\theta = \frac{1}{2}$
M1	$2\theta = 1.24(90...,)\ 4.39(06...)$
A1	$\theta = 0.625,\ 2.2(0)$
M1	$2\theta = 0.5235...,\ 2.61(79...)$ allow $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$
A1	$\theta = 0.262,\ 1.31$ allow $\left[\frac{\pi}{12}, \frac{5\pi}{12}\right]$

Type MS · Proving identities with double-angle and addition formulae

Answer 7 of 14 · Question 8 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Proving identities with double-angle and addition formulae

Answer 8 of 14 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

4. Use of calculators

Unless the question specifically states 'show' or 'prove' accept correct answers from no working. If an incorrect solution is given without any working do not award the Method mark.

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Proving identities with double-angle and addition formulae

Answer 8 of 14 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this: eg in a case of "prove or show....")

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Multiple attempts at a question.

If there is a choice of methods shown, then award the lowest mark, unless the subsequent working makes clear the method that has been used.

Type MS · Proving identities with double-angle and addition formulae

Answer 8 of 14 · Question 1 part b · 3 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
1 (a)	$\lfloor \cos 2A = \cos^2 A - \sin^2 A \quad \text{and} \quad \cos^2 A + \sin^2 A = 1 \rfloor$ $\Rightarrow [\cos 2A] = (1 - \sin^2 A) - \sin^2 A$ $\Rightarrow [\cos 2A] = 1 - 2\sin^2 A *$	M1 A1 cso [2]
(b)	$\frac{4-y}{3} = 1 - 2\left(\frac{x+1}{2}\right)^2$ $\Rightarrow \frac{4-y}{3} = 1 - 2\left(\frac{x^2+2x+1}{4}\right) \Rightarrow y = ...$ $\Rightarrow y = \frac{1}{2}(3x^2 + 6x + 5)$ *	M1 M1 A1* cso [3]
		Total 5 marks

Type MS · Proving identities with double-angle and addition formulae

Answer 9 of 14 · Question 11 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

11(a)	$\cos 2A = \cos^2 A - \sin^2 A \Rightarrow \cos 2A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ $2\sin^2 A = 1 - \cos 2A \Rightarrow \sin^2 A = \frac{1}{2}(1-\cos 2A)$ *	M1M1 A1cso [3]
(b)	$\cos^2 A = \frac{1}{2}(1+\cos 2A)$ $\sin^4 x + \cos^4 x = \left(\left(\frac{1-\cos 2x}{2}\right)^2 + \left(\frac{1+\cos 2x}{2}\right)^2\right)$ $= \frac{1}{4}\left((1-\cos 2x)^2 + (1+\cos 2x)^2\right)$ $= \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x + 1 + 2\cos 2x + \cos^2 2x\right)$ $= \frac{1}{4}\left(2 + 2\cos^2 2x\right) = \left\{\frac{1}{2}\left(1+\cos^2 2x\right)\right\}$ $\left\{\cos^2 2x = \frac{1+\cos 4x}{2}\right\}$ $\sin^4 x + \cos^4 x = \frac{1}{2}\left(1 + \frac{1+\cos 4x}{2}\right) = \frac{3+\cos 4x}{4}$ *	B1 M1 A1 M1A1 cso [5]
(c)	$5\sin 2\theta + 6 = 8\left(\frac{3+\cos 2\theta}{4}\right) \Rightarrow 5\sin 2\theta + 6 = 6 + 2\cos 2\theta$ $\Rightarrow 5\sin 2\theta = 2\cos 2\theta \Rightarrow \frac{\sin 2\theta}{\cos 2\theta} = \frac{2}{5} \Rightarrow \tan 2\theta = \frac{2}{5}$ $2\theta = 21.801...^\circ, 201.801...^\circ, 381.801...^\circ$ $\Rightarrow \theta = 10.9^\circ, 100.9^\circ$ Penalise extra angles in range by withholding the final A mark. Extra angles out of range – ignore.	M1 M1 A1 A1 [4]
		Total 12 marks

Part	Mark	Notes
(a)	M1	For using the summation formula for cos 2A They must start with either $\cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A \cos A - \sin A \sin A$
	M1	For eliminating $\cos^2 A$ using $\cos^2 A + \sin^2 A = 1$ and attempting to rearrange to the required result. This is not dependent on the first M mark, so if they start with another identity for cos2A they can still get this mark.
	A1 cso	For the correct identity with no errors seen. This is a given result.
	NB Some candidates work backwards – that is fine, please follow their working.	

Type MS · Proving identities with double-angle and addition formulae

Answer 9 of 14 · Question 11 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Working with a different variable. If they work in this part with a different variable (eg A) then award all the marks as appropriate up to the last mark. If they leave their final answer in terms of another variable, withhold the final A mark only. If they however, change to x on the final line award all the marks [provided everything is correct]		
(b)	**Main method**	
	B1	For use of the correct identity for $\cos^2 x$ $\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$ or if they convert $\cos^2 x$ to $\sin^2 x$ use the Pythagorean identity $\cos^2 x = 1-\sin^2 x$ and apply the given identity.
	M1	For squaring both identities. This must be a correct expansion. Ft their identity for $\left[\cos^2 x = \frac{1}{2}(1+\cos 2x)\right]$
	A1	For collecting like terms and obtaining $\frac{1}{2}(1+\cos^2 2x)$ oe. For example, $\frac{1}{4}(2+2\cos^2 2x)$
	M1	For applying the identity: $\cos^2 A = \frac{1}{2}(1+\cos 2A) \Rightarrow \left[\cos^2 2A = \frac{1}{2}(1+\cos 4A)\right]$ on $\cos^2 2x$ **only** again to achieve an expression in $\cos 4x$ **only**
	A1 cso	For the correct identity with no errors seen. This is a given result. You must check every line of their work carefully.
	ALT 1	
	B1	For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working].
	M1	For using the expansion of $\sin^2 x+\cos^2 x$ as follows $\sin^4 x+\cos^4 x = \left(\sin^2 x+\cos^2 x\right)^2 - 2\sin^2 x\cos^2 x$ This must be correct
	A1	For obtaining $\sin^4 x+\cos^4 x = 1-\frac{1}{2}\sin^2 2x$ oe. For example; $\frac{1}{2}(2-\sin^2 2x)$
	M1	For applying the given identity on $\sin^2 2x$ **only** to achieve an expression in $\cos 4x$ **only** $\sin^2 A = \frac{1}{2}(1-\cos 2A) \Rightarrow \left[\sin^2 2A = \frac{1}{2}(1-\cos 4A)\right]$ $\sin^4 x+\cos^4 x = 1-\frac{1}{2}\left(\frac{1-\cos 4x}{2}\right) \Rightarrow \left(\sin^4 x+\cos^4 x = \frac{3+\cos 4x}{4}\right)$
	A1 cso	For the correct identity with no errors seen. This is a given result. You must check every line of their work carefully.

Type MS · Proving identities with double-angle and addition formulae

Answer 9 of 14 · Question 11 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	ALT 2 – Works backwards from the given result	
	B1	For use of the correct identity for $\sin 2A = 2\sin A\cos A$ [seen later in their working].
	M1	Applies the cos2A identity and converts 3 into $3\left(\sin^2 2x+\cos^2 2x\right)$ $\frac{3+\cos 4x}{4}=\frac{3\sin^2 2x+3\cos^2 2x+\left(\cos^2 2x-\sin^2 2x\right)}{4}$
	A1	Obtains $\frac{3+\cos 4x}{4}=\frac{4\cos^2 2x+2\sin^2 2x}{4}$
	M1	Applies the cos2A identity and expands the bracket. $\frac{3+\cos 4x}{4}=\frac{4\left(\cos^2 x-\sin^2 x\right)^2+8\sin^2 x\cos^2 x}{4}$ $=\frac{4\cos^4 x-8\sin^2 x\cos^2 x+4\sin^4 x+8\sin^2 x\cos^2 x}{4}$
	A1 cso	Simplifies to the required result with no errors seen $\frac{3+\cos 4x}{4}=\sin^4 x+\cos^4 x$ *

(c)	M1	For obtaining the correct equation in terms of $\sin 2\theta$ and $\cos 2\theta$ $6+2\cos 2\theta=5\sin 2\theta+6$ Accept unsimplified, accept even: $8\left(\frac{3+\cos 2\theta}{4}\right)=5\sin 2\theta+6$
	M1	For using the $\tan 2\theta=\frac{\sin 2\theta}{\cos 2\theta}$ identity correctly on their expression following an expression in the form $A\cos 2\theta=B\sin 2\theta$ and must be in terms of 2θ
	A1	For achieving at least one correct angle for 2θ **NB This is an M mark in Epen**
	A1	For awrt both 10.9° and 100.9° Penalise extra angles in range by withholding the final A mark. Extra angles out of range – ignore.

Type MS · Proving identities with double-angle and addition formulae

Answer 9 of 14 · Question 11 part b · 5 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

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Type MS · Proving identities with double-angle and addition formulae

Answer 10 of 14 · Question 10 part c · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

10 (a)	$f\left(\frac{3}{4}\right)=16\left(\frac{3}{4}\right)^3+11\left(\frac{3}{4}\right)-15$ * $\left(\frac{27}{4}+\frac{33}{4}-15\right)=0$	M1 A1cso [2]
(b)(i)	$[\sin 2\theta=\sin(\theta+\theta)]=\sin\theta\cos\theta+\sin\theta\cos\theta$	M1
	$(\sin 2\theta=)2\sin\theta\cos\theta$ *	A1cso [2]
(b)(ii)	$[\cos 2\theta=\cos(\theta+\theta)=]\cos\theta\cos\theta-\sin\theta\sin\theta=\cos^2\theta-\sin^2\theta$	M1
	$=\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$ *	M1 A1cso [3]
(c)	$27\cos\theta(2\cos^2\theta-1)+19\sin\theta(2\sin\theta\cos\theta)-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38(1-\cos^2\theta)\cos\theta-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38\cos\theta-38\cos^3\theta-15(=0)$ $(16\cos^3\theta+11\cos\theta-15=0)$ oe	A1
	(When $x=\cos\theta$) $16x^3+11x-15=0$ *	A1cso [4]
(d)	$(16x^3+11x-15=0\Rightarrow(4x-3)(4x^2+3x+5)=0)\rightarrow 4x^2+3x+5$	M1
	$b^2-4ac=9-80<0$ (So no real roots)	M1 A1
	(Only solution is) $4x-3=0$ or $4\cos\theta-3=0$ so $\cos\theta=\frac{3}{4}$ *	A1cso [4]
		Total 15 marks

Type MS · Proving identities with double-angle and addition formulae

Answer 10 of 14 · Question 10 part c · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Proving identities with double-angle and addition formulae

Answer 11 of 14 · Question 10 part b · 4 marks

2023 NOV · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
10(a)(i) (ii)	$\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A$ *	B1 cso
	$\cos(A+A)=\cos A\cos A-\sin A\sin A$	
	$\Rightarrow \cos 2A=\cos^2 A-\sin^2 A$	M1
	$=\cos^2 A-(1-\cos^2 A)$	A1*cso
	$=2\cos^2 A-1$ *	[3]
(b)	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos^2\theta}-\frac{\cos^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}\times\frac{\cos^2\theta}{\cos^2\theta}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$	dM1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)$ or $\left(\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
ALT	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{1+\tan^2\theta}$	dM1
	$\cos^2\theta+\sin^2\theta=1\Rightarrow\tan^2\theta+1=\frac{1}{\cos^2\theta}\Rightarrow\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
(c)	$\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)}=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)\Rightarrow\frac{5}{2}\sin\left(2\left(x+\frac{\pi}{6}\right)\right)=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$	M1
	$1-2\cos^2\left(x+\frac{\pi}{6}\right)=-\cos\left(2x+\frac{\pi}{3}\right)$	M1
	$\left(\frac{5}{2}\sin\left(2x+\frac{\pi}{3}\right)=-\cos\left(2x+\frac{\pi}{3}\right)\Rightarrow\right)\tan\left(2x+\frac{\pi}{3}\right)=-\frac{2}{5}$	M1
	$\left(\Rightarrow 2x+\frac{\pi}{3}=\right)-0.38050,\quad 2.76108$	M1
	$(\Rightarrow x=)-0.714,\quad 0.857$	A1A1 [6]

Type MS · Proving identities with double-angle and addition formulae

Answer 11 of 14 · Question 10 part b · 4 marks

2023 NOV · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

(d)	$I = \int_0^{\frac{\pi}{2}} (2\sin 2\theta - \cos 5\theta + 2)\ \{\mathrm{d}\theta\}$	M1
	$= \left[\frac{-2\cos 2\theta}{2} - \frac{\sin 5\theta}{5} + 2\theta\right]_0^{\frac{\pi}{2}}$	M1
	$= \left(-\cos 2\left(\frac{\pi}{2}\right) - \frac{\sin 5\left(\frac{\pi}{2}\right)}{5} + 2\times\left(\frac{\pi}{2}\right)\right) - \left(-\cos 2(0) - \frac{\sin 5(0)}{5} + 2\times 0\right)$	M1
	$= \left(1 - \frac{1}{5} + \pi\right) - (-1 - 0 + 0) = \frac{9}{5} + \pi$	A1 [4]
		Total 17 marks

Type MS · Proving identities with double-angle and addition formulae

Answer 11 of 14 · Question 10 part b · 4 marks

2023 NOV · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1 cso	For $\sin(A+A)$ or $\sin 2A = \sin A\cos A + \cos A\sin A = 2\sin A\cos A$ * No steps missing or incorrect work. Condone $\sin A + A$ with brackets missing if all other work correct.
	M1	For $\cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A$
	A1* cso	For $\cos(A+A)$ or $\cos 2A = \cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A)$ $= 2\cos^2 A - 1$ No steps missing, no incorrect work. Condone $\cos A + A$ with brackets missing if all other work correct.
Students who miss 'A' in their working, (eg coscos – sinsin) may be awarded M mark, but not A or B marks.		
(b)	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. Allow the 2 to be missing for this mark **but not the final A mark.**
	dM1	For $1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$ or multiplying by $\frac{\cos^2\theta}{\cos^2\theta}$ to give $\left(\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \right) \frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ The $\times\frac{\cos^2\theta}{\cos^2\theta}$ does not need to be explicitly written. Dependent on previous method mark.
	ddM1	For minimum steps (and dependent on previous method marks): $\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta} = 2\sin\theta\cos\theta$
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
ALT	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity.
	dM1	For replacing $1 + \frac{\sin^2\theta}{\cos^2\theta}$ with $1 + \tan^2\theta$ Dependent on previous method mark.
	ddM1	For replacing $\tan^2\theta + 1 = \frac{1}{\cos^2\theta} \Rightarrow \frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}} = 2\sin\theta\cos\theta$ Dependent on both previous method marks.
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
Students who miss θ in their working, may be awarded M marks, but not the A mark.		

Type MS · Proving identities with double-angle and addition formulae

Answer 11 of 14 · Question 10 part b · 4 marks

2023 NOV · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Allow work throughout where candidates have let, for example $\theta = x + \frac{\pi}{6}$ and worked in θ and 2θ throughout in (c)		
(c)	M1	Rearranges the given equation into $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)} = \left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ **and** changes the left hand side into $P\sin\left(2\left(x+\frac{\pi}{6}\right)\right)$ where P is any constant.
	M1	Correctly states or uses $1-2\cos^2\left(x+\frac{\pi}{6}\right) = -\cos\left(2x+\frac{\pi}{3}\right)$ (condone being replaced with $-\cos 2A$)
	M1	For $\tan\left(2x+\frac{\pi}{3}\right) = Q$ where Q is any constant.
	M1	For -0.38050 or 2.76108 This may be awarded for either value being seen, ignore any extra values seen. Accept any values with at least 3 significant figures.
	A1	For $(x=)-0.714$ or $(x=)0.857$ This may be awarded for either value being seen, ignore any extra values seen. Accept any value with more than 3 significant rounding to those given.
	A1	For $(x=)-0.714$ and $(x=)0.857$ Ignore any extra values **out of range**. Extra values in range will mean A0. Accept any value with more than 3 significant rounding to those given.
Students must use θ or $x+\frac{\pi}{6}$ and 2θ or $2\left(x+\frac{\pi}{6}\right)$		
(d)	M1	For changing the integral into $(I=)\int_0^{\frac{\pi}{2}}(2\sin 2\theta - \cos 5\theta + 2)\,\{\mathrm{d}\theta\}$ $\mathrm{d}\theta$ may be omitted. If the integral sign and limits are not present, these can be implied by a clear attempt in later work to integrate all the terms and a substitution of limits. Students must use 2θ and 5θ in their working.
	M1	For a minimum of 2 out of the 3 terms shown above correctly integrated. Students must use 2θ and 5θ in their working.
	M1	For substituting in the given limits the correct way around into their changed function with at least two terms, there must be at least one explicit substitution of each limit. If the candidate has a constant term as part of the function, they can omit this from this step (as the terms cancel out anyway). This mark may be implied by a correct answer. If the candidate is working with an incorrect function and substitution of 0 clearly gives 0, this substitution does not need to be shown.
	A1	For $\frac{9}{5}+\pi$ The approximate decimal value 4.941…will **usually** indicate correct working (as a guide), but a substitution of limits will be needed to get the 3rd method mark.
In this question, students are directed to use calculus, so must show a correct integration. The penultimate method mark may be implied from a fully correct answer, but if the answer is not correct, we must see substitution of limits as stated.		

Type MS · Proving identities with double-angle and addition formulae

Answer 11 of 14 · Question 10 part b · 4 marks

2023 NOV · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
	As with many log questions, there are a number of ways of reaching a correct answer. **The first scheme shows 3 common routes, with a set of general principles in the notes.** **Not all variations in methods will have been captured, please mark to general principles, referring ANY work where this is not possible and worthy of marks to REVIEW.** **This includes work that converts everything to completely new bases.**	

Type MS · Proving identities with double-angle and addition formulae

Answer 12 of 14 · Question 11 part a · 4 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

11 (a)	$\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$	M1
	$=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ *	M1M1A1 cso
	ALT example	[4
	$\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$	
	$=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$	2nd M1
	$=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$	1st M1
	$=(1-2\cos^2\theta)$	3rd M1
	$=\cos2\theta$	A1
(b)	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$	M1
	$=2\cos\left(2\theta+\frac{\pi}{4}\right)$	A1
	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$	M1
	$\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$	
	$\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$	A1
	$2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$	dM1
	$\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$	A1A1 [7]
(c)	$\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos4x-8\sin4x\right)\mathrm{d}x$	M1
	$=\left[\frac{1}{4}\sin4x+\frac{8}{4}\cos4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$	M1
	$=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$	M1
	$=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$	A1 [4]
		Total 15 marks

Type MS · Proving identities with double-angle and addition formulae

Answer 12 of 14 · Question 11 part a · 4 marks

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Part	Mark	Additional Guidance
General Principles for marking part (a) • First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms. • Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen. • Third M mark is for applying any identity for $\cos 2\theta$ • A1 For obtaining the required result with no errors.		
(a)	M1	For factorising as the difference of two squares $(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$
	M1	For use of $\cos^2\theta+\sin^2\theta)=1$
	M1	For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$
	A1 cso	Obtain the given result with no errors in the working.
(b)	M1	For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	A1	For rearranging to the correct quadratic
	M1	For a valid attempt to solve their **3TQ**. See General Guidance for what constitutes an attempt to factorise. If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions.
	A1	For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	dM1	For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$ i.e. 0.7227…or $\frac{2}{3}\pi$ Accept values out of range for this mark. This is dependent on the previous M mark.
	A1	For two correct solutions in range
	A1	For all four correct solutions in range and no others. Extra solutions within range scores A0 Ignore solutions outside of range. Exact solutions when given must be given as multiples of π.
(c)	M1	For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$
	M1	For an attempt to integrate $\cos kx\rightarrow l\sin kx$ where $k,l\neq 1$
	M1	For substituting limits into a changed expression
	A1	For the correct answer in the required form.

Type MS · Proving identities with double-angle and addition formulae

Answer 12 of 14 · Question 11 part a · 4 marks

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Type MS · Proving identities with double-angle and addition formulae

Answer 13 of 14 · Question 11 part b · 3 marks

2020 JAN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

11(a)	$(3\sin A\cos B - 3\cos A\sin B) = (\sin A\cos B + \cos A\sin B)$	M1
	$2\sin A\cos B = 4\cos A\sin B$	M1
	$\rightarrow \frac{\sin A}{\cos A} = 2\frac{\sin B}{\cos B}$	M1
	$\tan A = 2\tan B \quad k = 2$	A1 (4)
(b)	$\frac{(\cos^4\theta - \sin^4\theta)}{\cos^2\theta} = \frac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= \frac{(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= 1 - \tan^2\theta$ *	A1 cso (3)
ALT 1	$1-\tan^2\theta = 1 - \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta}$	M1
	$= \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta} \times \left(\cos^2\theta + \sin^2\theta\right)$	M1
	$= \frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta}$	A1
ALT 2	$\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} = \cos^2\theta - \frac{\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \tan^2\theta\sin^2\theta$	M1 Eliminate 4th powers
	$= \cos^2\theta - \tan^2\theta\left(1 - \cos^2\theta\right)$	M1 Eliminate $\sin^2$
	$= \cos^2\theta - \tan^2\theta + \sin^2\theta = 1 - \tan^2\theta$	A1
(c)(i)	$\cos(45-30)$ or $\cos(60-45) = \frac{1}{2}\times\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}$	M1
	$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \frac{\sqrt{2}+\sqrt{6}}{4}$ *	A1cso (2)
ALT	By using double angle formula: $\cos^2 15° = \frac{1}{2}\left(1+\cos 30°\right) = \frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)$	M1
	Leading to the *given* answer. $\cos 15° = \sqrt{\left(\frac{2+\sqrt{3}}{4}\right)}$ or $\frac{\sqrt{2+\sqrt{3}}}{2}$ must be seen.	A1

Type MS · Proving identities with double-angle and addition formulae

Answer 13 of 14 · Question 11 part b · 3 marks

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Question Number	Scheme	Marks
(ii)	$\tan 255 = \tan 75$	B1
	$= \tan(30+45) = \dfrac{\tan 30 + \tan 45}{1 - \tan 30 \tan 45}$	M1
	$= \dfrac{\frac{\sqrt{3}}{3}+1}{1-\frac{\sqrt{3}}{3}}$	dM1
	$\dfrac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \dfrac{3+\sqrt{3}}{3-\sqrt{3}}$ *	A1cso (4)
		[13]
(a)		
M1	Expand both sides of the equation using correct formulae	
M1	Collect like terms from their expansions. (Not dependent)	
M1	Divide through by $\cos A \cos B$	
A1	Replace each fraction with the appropriate tangent and show $k = 2$ (value need not be shown explicitly)	
(b)		
M1	Factorise the numerator using the difference of 2 squares.	
M1	Replace $\sin^2\theta + \cos^2\theta$ with 1	
A1cso	Divide both terms by $\cos^2\theta$ and obtain the *given* answer with no errors seen.	
ALT 1		
M1	Use $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ and obtain a single fraction with no tan	
M1	Indicate multiplication by $\sin^2\theta + \cos^2\theta$	
A1cso	Multiply and obtain the *given* answer with no errors seen.	
(c)(i)		
M1	Express 15 as the difference of 2 suitable numbers, expand using a correct formula and substitute the correct exact values for the trig functions (substitution must be shown).	
A1cso	Simplify and combine the fractions to obtain the *given* answer with no errors seen.	
(ii)		
B1	$\tan 255 = \tan 75$ seen explicitly or used. OR eg tan(210 + 45) – give B1 for tan 210 = tan 30 used	
M1	Express 75 as 30 + 45 and expand $\tan(30+45)$ using the correct formula (given on the formula page) OR expand tan (210 + 45) If 75 = 15 + 60 is used tan 15 can be obtained from a calculator but must be in exact form..	
dM1	Substitute the correct exact values for the trig functions	
A1cso	Simplify the fractions to obtain the *given* answer with full working and no errors seen.	

Type MS · Proving identities with double-angle and addition formulae

Answer 13 of 14 · Question 11 part b · 3 marks

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Type MS · Proving identities with double-angle and addition formulae

Answer 13 of 14 · Question 11 part b · 3 marks

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Type MS · Proving identities with double-angle and addition formulae

Answer 14 of 14 · Question 11 part b · 3 marks

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11(a)	$cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ $\Rightarrow cos\,2A = cos^2 A - (1 - cos^2 A)$ $cos\,2A = 2\,cos^2 A - 1$ *	M1 A1 cso [2]
(b)	$(2\cos^2 A - 1)^2 = (\cos 2A)^2$ $\cos 4A = 2\cos^2 2A - 1 \Rightarrow \cos^2 2A = \frac{\cos 4A + 1}{2}$ $(2\cos^2 2A - 1)^2 = \frac{\cos 4A + 1}{2}$ *	M1 M1 A1 cso [3]
(b) **ALT**	$cos\,4A = 2\,cos^2 2A - 1$ $= 2(2\,cos^2 A - 1)^2 - 1$ $(2\,cos^2 A - 1)^2 = \frac{cos\,4A + 1}{2}$ *	M1 M1 A1 cso [3]
(c)	$y = \frac{sin\,2x}{2} + \frac{(2\,cos^2 x - 1)^2}{2} + \frac{1}{8} \Rightarrow y = \frac{sin\,2x}{2} + \frac{cos\,4x}{4} + \frac{3}{8}$ $\frac{dy}{dx} = cos\,2x - sin\,4x$ $\frac{dy}{dx} = cos\,2x - 2\,sin\,2x\,cos\,2x \{= 0\}$ $\{cos\,2x(1 - 2\,sin\,2x) = 0\} \Rightarrow sin\,2x = \frac{1}{2}$ $\Rightarrow x = \frac{\sin^{-1}\left(\frac{1}{2}\right)}{2} = \cdots$ $x = \frac{\pi}{12}$ $y = \frac{sin2\left("\frac{\pi}{12}"\right)}{2} + \frac{cos4\left("\frac{\pi}{12}"\right)}{4} + \frac{3}{8} = \frac{3}{4} \{\Rightarrow \left(\frac{\pi}{12}, \frac{3}{4}\right)\}$	M1 dM1 B1 ddM A1 M1A1 [7]
		Total 12 marks

Type MS · Proving identities with double-angle and addition formulae

Answer 14 of 14 · Question 11 part b · 3 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correct use of $cos(A+B) = cos\,A\,cos\,B - sin\,A\,sin\,B \Rightarrow$ $cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ ***OR*** $cos\,2A = cos^2 A - sin^2 A$ **with** the Pythagorean identity to eliminate sine from the identity.
	A1 cso	Achieves printed answer with no errors seen. Left hand side must be cos2A not cos(A+A)
(b)	M1	Uses the given result from (a), replaces $(2\cos^2 A - 1)^2$ with $(\cos 2A)^2$
	M1	Uses the given result from (a) or a correct identity to replace $\cos^2 2A$ with $\frac{\cos 4A+1}{2}$
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(b) ALT	M1	Uses the given result from (a) on cos 4A
	M1	Uses the given result from (a) or a correct identity on cos 2A
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(c)	M1	Rearranges the given y into the form $\frac{\sin 2x}{2} + \frac{\cos 4x+1}{k} + l$, oe, where k and l are nonzero constants Or $\frac{\sin 2x}{2} + \frac{\cos 4x}{P} + Q$, oe, where P and Q are nonzero constants (Allow mixed variables x and A for this method mark only)
	dM1	For an attempt to differentiate their y into the form $P\cos(2x) \pm Q\sin(4x)$ *oe, where P, Q are nonzero constants* **Depends on the previous method mark.**
	B1	Uses $\sin 4x = 2\sin 2x\cos 2x$ or $\sin 4x = \sqrt{1-\cos^2 4x}$ oe in their dy/dx (may be implied by correct working)
	ddM1	Sets their differentiated expression = 0 (can be implied) and attempts a correct method to solve for x=... **Depends on both previous method marks.**
	A1	For $x = \frac{\pi}{12}$, ignore extra x values are outside of given range. Withhold A mark if extra solution seen in the given range.
	M1	For attempting to find the value of y using their x, found by solving their $\frac{dy}{dx} = 0$, provided x is in the range $0 \leqslant x \leqslant \frac{\pi}{6}$ Substitutes into the given y **or** their y, correct answer implies this mark, but for incorrect answer, must see explicit substitution
	A1	For the correct exact coordinates of P, $\left(\frac{\pi}{12}, \frac{3}{4}\right)$ *or* $\left(\frac{\pi}{12}, 0.75\right)$, accept $x = \frac{\pi}{12}$, $y = \frac{3}{4}$ *or* $x = \frac{\pi}{12}$, $y = 0.75$

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